ABHIJITH SIVAPRASADAN

M.Sc. Sustainable Energy Engineering, KTH 2026 | PyPSA Grid Flow Modelling | BESS & Grid Flexibility | Python Decision-Support Tools

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PROFILE

Energy systems modeller completing an M.Sc. in Sustainable Energy Engineering at KTH, focused on PyPSA grid-flow modelling, renewable integration, BESS/flexibility analytics, Python optimisation and decision-support tools. Built a dedicated Netherlands-inspired PyPSA-NL grid flexibility platform around congestion, connection-capacity limits, curtailment, BESS siting/sizing, flexible connection contracts, bottleneck diagnostics and N-1 screening proxies. Combines power/energy-systems coursework, Python data/software delivery, and industrial R&D validation experience from Siemens Energy and Alleima. Comfortable building models from scratch, validating outputs, and translating technical results into dashboards, reports and commercial/engineering insights.

EXPERIENCE

Test Engineer / Master Thesis Student

Siemens Energy AB, Fluid Dynamic Lab, Finspang — Apr 2025 - Nov 2025

- Built steady-state compressible CFD and conjugate heat-transfer models in ANSYS Fluent for high-temperature test-rig hardware, using consistent boundary conditions, engineering KPIs, mesh/near-wall checks and sensitivity studies.
- Translated simulation outputs into design recommendations and validation priorities, documenting assumptions, uncertainty drivers and next-step experimental verification needs.
- Commissioned and verified NI-DAQ measurement chains involving thermocouples, pressure sensors, NI MAX and LabVIEW acquisition workflows; strengthened data-quality, troubleshooting and test-readiness discipline.

Energy Efficiency Intern

Alleima AB, Sandviken (Remote) — Dec 2024 - Jun 2025

- Developed an ISO 50001/EED-aligned energy performance mapping approach for industrial energy systems, including KPI/EnPI definitions, baselines, load-driver normalisation and measurement-readiness review.
- Built analytical workflows to compare operating states, identify loss mechanisms and anomalies, and support energy-efficiency improvement roadmaps across utilities/process-energy contexts.

Backend Engineer

QBurst, Thiruvananthapuram, India — Aug 2021 - Jul 2023

- Developed and maintained backend APIs, automated tests and data-handling workflows using Python/TypeScript, Git, Postman and Docker in agile software teams.
- Built transferable experience in structured debugging, maintainable code, reproducible workflows, version control, technical documentation and cross-functional delivery.

RELEVANT PROJECTS

PyPSA-NL Grid Congestion, BESS & Flexible Connection Modelling Platform

Portfolio Project | Python, PyPSA, pandas, Streamlit, Plotly, Linopy, HiGHS — 2026

GitHub: github.com/abhijith-sivaprasadan/pypsa-nl-grid-flexibility

- Built a Netherlands-inspired constrained electricity-grid modelling platform in PyPSA to analyse congestion, connection-capacity limits, renewable curtailment, BESS flexibility, flexible connection contracts and targeted grid reinforcement.
- Modelled a simplified regional grid with provincial buses, inter-regional corridors, hourly demand, solar PV, onshore wind, offshore wind, backup generation and storage/flexibility options, using transparent CBS-calibrated demand and renewable-capacity assumptions.
- Developed scenario workflows covering base constrained grid, high solar/wind growth, regional BESS placement, flexible connection contracts, targeted reinforcement and combined BESS/flexibility cases.
- Used PyPSA/Linopy/HiGHS optimisation to solve hourly dispatch under grid constraints and compare objective cost, renewable dispatch, backup generation, line loading, curtailment and emissions-proxy outcomes.
- Implemented BESS siting/sizing sweeps across candidate Dutch regions and storage durations, ranking options by renewable dispatch gain, backup reduction, congestion-cost proxy relief, line-overload reduction and utilisation metrics.
- Built bottleneck diagnostics, N-1 screening proxy, validation checks, automated CSV outputs, executive summaries and a Streamlit dashboard for scenario comparison, hourly flows, BESS ranking, validation review and dynamic interpretation.
- Latest generated run passed 8/8 validation checks; top scenario reduced the congestion-cost proxy by about EUR 8.0M versus base, while the best BESS sweep option was Groningen, 400 MW / 4 h.

Distribution Grid Studies - EV Charging, PV and Storage Integration

KTH Course Project | Python / power-flow workflow — 2025

- Analysed a CIGRE MV distribution grid using load-flow, N-1 contingency analysis and 24-hour time-series simulation with EV charging, PV/storage integration, voltage stability and line-loading constraints.

District Heating Dispatch Optimisation

KTH Course Project | Python, PuLP, MILP — 2025

- Implemented a mixed-integer dispatch model comparing cost/emissions objectives, CHP and heat-pump operation, storage integration, outages, electricity-price coupling and future 2030 scenarios.

Day-Ahead Building Heating Demand Forecasting

KTH Course Project | Python, scikit-learn — 2025

· Built heating-demand forecasting models using weather and temporal features, comparing baseline methods and neural networks with structured feature engineering and model evaluation.

EDUCATION

M.Sc. in Sustainable Energy Engineering (expected 2026)

KTH Royal Institute of Technology, Stockholm 2023 - 2026

Relevant coursework: Practical Optimization of Energy Networks, Modelling of Energy Systems, Energy Management, AI Applications in Sustainable Energy Engineering, Sustainable Power Generation, Energy Systems for Sustainable Development, Numerical Heat Transfer in Energy Technology, Applied Heat and Power Technology.

Circular Economy for Energy Storage

Aalto University / Unite! Virtual Exchange Oct 2024 - Dec 2024

B.Tech in Mechanical Engineering

APJ Abdul Kalam Technological University, Kerala, India 2017 - 2021

Relevant coursework: Thermodynamics, Heat and Mass Transfer, Fluid Mechanics, Thermal Engineering, Compressible Flow, CAD/CAE and Mechanical Design.

TECHNICAL SKILLS

Grid & flexibility modelling: PyPSA, load-flow studies, N-1 screening/proxy, line loading, congestion, connection capacity, curtailment, BESS siting/sizing, solar/wind integration, flexible connection contracts, scenario analysis.

Optimisation & analytics: Linopy, HiGHS, PuLP/MILP, dispatch optimisation, cost/emissions objective formulation, sensitivity analysis, KPI design, validation checks, techno-economic analysis.

Programming, data & dashboards: Python, pandas, NumPy, SciPy, scikit-learn, Streamlit, Plotly, matplotlib, Git, Docker, CI/reproducible repositories, REST APIs, Postman.

Energy systems: Power systems, district heating dispatch, heat pumps and thermal systems, industrial energy performance mapping, ISO 50001/EED-style baselines and load-driver normalisation.

Simulation & testing: ANSYS Fluent, CFD/CHT, ANSYS Meshing, IDA ICE, HOMER Pro, SAM, NI-DAQ, NI MAX, LabVIEW, thermocouples, static/dynamic pressure sensors, measurement-chain verification.

CERTIFICATIONS

SSG Entre / Industrial Safety, Sweden | Applied Computational Fluid Dynamics, Siemens | IBM Python for Data Science | Digital Manufacturing & Design Technology, University at Buffalo | Schneider Electric - Assessing a Commercial Product’s Sustainability

LANGUAGES

English: professional pro
ficiency | Malayalam: native | Swedish: beginner, actively studying